

Anoop Deekshith R

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EDUCATION

University of California, Santa Cruz

Santa Cruz, CA

M.S in Electrical & Computer Engineering

September 2025 – June 2027 (exp)

Courses taken : Computer Architecture, VLSI Design, ASIC Design, Advance Microprocessor Design, FPGA Design

Visvesvaraya Technological University

Bengaluru, KA

B.E in Electronics & Communication Engineering GPA: 3.45

August 2017 – August 2021

SKILLS

Programming Languages: Python, C, C++, Verilog, VHDL, System Verilog, TCL basics

Open Source Tools: EDA Playground, Icarus Verilog, Gem5, openROAD, Linux, Github

Frontend Industry Tools: CUDA Basics, Cadence Xcelium Logic Simulator, Genus Synthesis Solution (basics)

Backend Industry Tools: Cadence Innovus, Tempus (basics), CAD Tools, LVS, DRC

Automated Test Equipment: Eagle Teradyne 300, Teradyne 88, Teradyne 364, T2000k, 93k

Statistical tools: Microsoft Word, Excel, Matlab (basics).

Lab Equipments: Voltage function generator, Oscilloscope, Probers, Soldering gun.

WORK EXPERIENCE

Anora Semiconductor Labs

Bengaluru, KA

Post Silicon validation Engineer - II

October 2021 - August 2025

- With 4 years of VLSI industry experience, developed and executed post-silicon validation and production test programs (FT & Prober) covering Scan, BIST, DC/AC parametrics, EEPROM, and digital interfaces (I²C, SPI, Manchester), ensuring robust design-to-silicon correlation for mixed-signal and sensor SoCs.
- Bridged front-end RTL and ATE platforms by converting Verilog patterns and timing files into C++ (T2K/T2000), optimizing test logic and reducing test time by 10%, contributing to a 40% overall efficiency gain during tester platform migrations.
- Performed electrical and hardware validation including probe-card support, pad-level checks, spike analysis on supply/critical pins, voltage/current compliance testing, and DRC/LVS-based pad verification, enabling reliable SoC bring-up and tape-out support.
- Led on-site validation and customer support in the Philippines, troubleshooting ETS88/T2K testers, debugging complex test flows on the production floor, executing yield correlation (Cp, Cpk, GRR) to sustain 99% yield, and ensuring smooth project transfer through strong client-facing collaboration.

Entuple Technologies

Bengaluru, KA

Physical Design Engineer Intern

April 2020 - June 2020

- Developed a solo project on an amplifier device by improving routing and placement, leveraging the knowledge acquired by the Physical Design methodology training.

SELECTED PUBLICATIONS

- (Accepted) **DEVELOPMENT OF PROSTHETIC HAND FOR THE DISABLED.** International Advanced Research Journal in Science, Engineering and Technology Vol. 8, Issue 5, May 2021
- (Process) **VOLTAGE DROOP DETECTION TECHNIQUES FOR ASIC'S: A Survey**, IEEE 2025

PROJECTS

Implementation of Best offset Prefetcher | RISC V, Scrab (UCSC's Architecture platform), Verilog, Gem5 September 2025

- Designed and implemented a Best-Offset (BO) Hardware Prefetcher in a cycle-accurate simulator (Scarab), and Gem5 achieving a 10.6% average IPC speedup across SPEC CPU2017 workloads, outperforming industry-standard Sandbox prefetching techniques.
- Optimized memory hierarchy performance by leveraging a Recent Requests (RR) table to prioritize prefetch timeliness over raw accuracy, resulting in a 52% average reduction in L1 cache misses.

Simple RISC V RV32I Core(Subset pipeline) | Verilog, UVM, Xcelium,RTL

October 2025

- Designing a 5-stage pipeline for a subset of the RV32I instruction set in Verilog. Planning on functional verification using UVM-based testbenches and Xcelium simulation, verifying arithmetic, control, and memory instructions. Documented RTL design, timing reports, and handling of pipeline hazards (on-going).

FPGA-based Counter & Memory Modules | FPGA, Xilinx, RTL, Verilog HDL

December 2025

- Synthesizing digital modules and deploying them on Xilinx/Intel FPGA, performing resource utilization and timing analysis. Planning to Optimize RTL for area and timing, achieving efficient clock frequency and low-latency memory access(on - going) .

GPU-Accelerated VHDL-Based Matrix Multiplier | FPGA, GPU, CUDA, Innovus

December 2025

- Designing a matrix multiplication engine in VHDL with pipelined and parallel execution inspired by GPU architecture and computer architecture concepts, targeting FPGA implementation for high-throughput computation (on-going).

Voltage Droop Prediction module | System on Chip, System Verilog, Innovus, Xcelium, Genus, TCL

September 2025

- Developed a deep learning-based voltage droop detection and prediction module to anticipate on-chip voltage droop events ahead of time, improving reliability in high-performance designs. Leveraged Convolutional Neural Networks (CNNs) to learn temporal and spatial features from power-grid measurement data, enabling early identification of droop patterns prior to their occurrence.(on-going)